

Calculus (PR12)

Week 1: Limits, Continuity and Derivatives

The Team



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Textbook

Calculus: A Complete Course

10th Edition



Weekly study chapters:

According to the syllabus published on
the Moodle!

Course Info

- 5 Study Points (15 Study Points = full-time = 40h)
- **Research Report + Presentation (30%)**
- 1 written exam (70%)



Note: Research Report + Presentation
instead of **Assignments + Elucidations**

Learning Goals



Limits

- Understand **limits** as the **foundation of calculus**
- Develop an intuitive understanding of **approaching a value**



Continuity

- Explain and identify **continuity at a point**
- Recognize and interpret discontinuities from graphs



Derivatives

- Compute derivatives as **rates of change** (tangent slope)
- Apply the **power rule** to compute derivatives **efficiently**



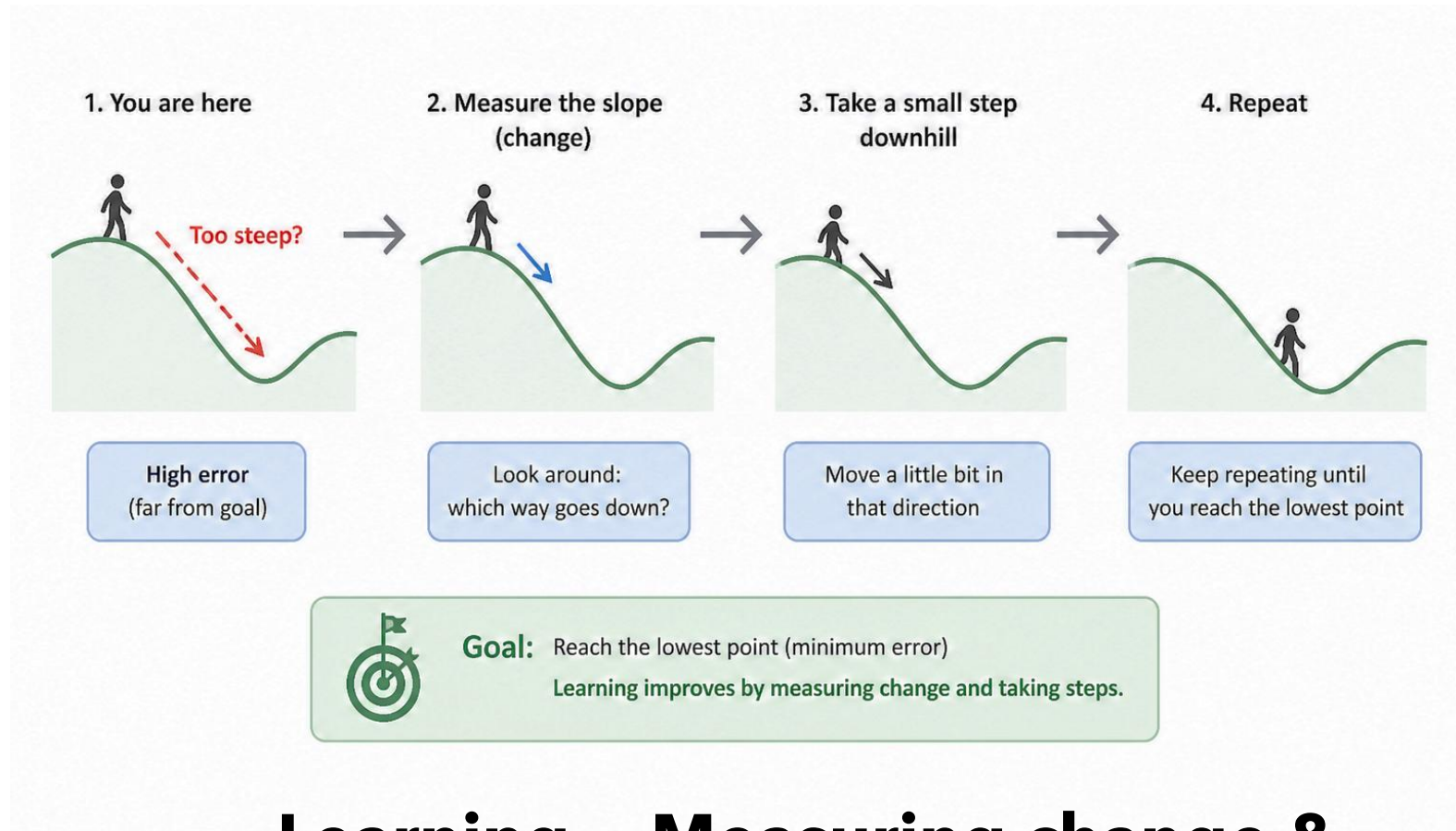
Why Calculus Matters?

Question:

What do you think is the connection between this video and calculus?

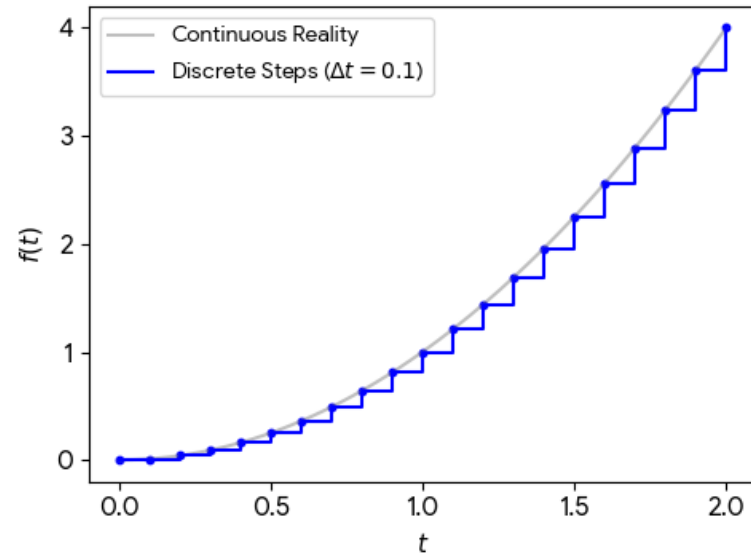
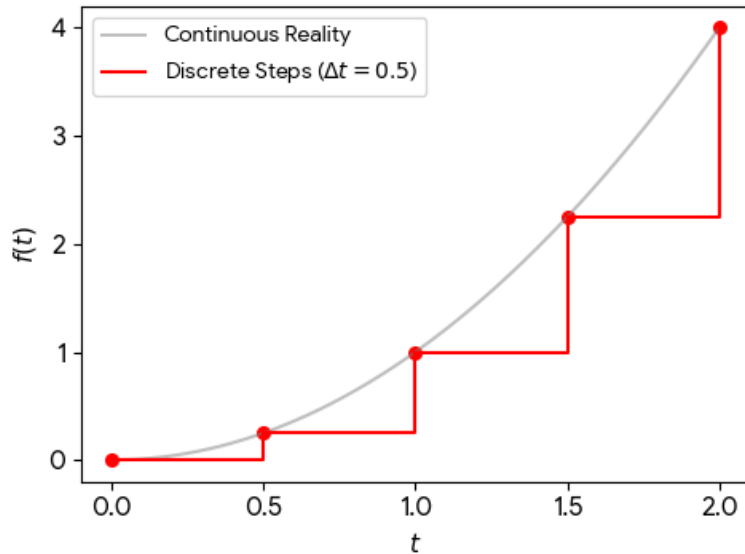
<https://svs.gsfc.nasa.gov/5632/>

Why Calculus Matters for AI?



Learning = Measuring change & Taking a Step

From Continuous Reality to Digital Understanding



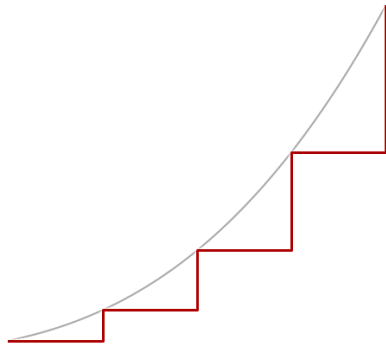
What is a Limit?

Definition:

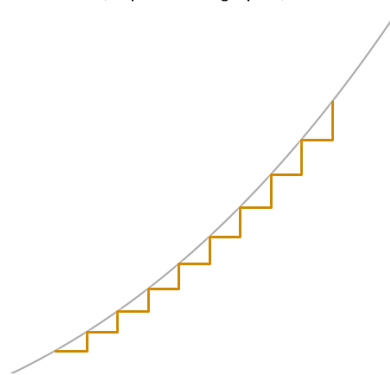
- A limit **asks what value $f(x)$ approaches** as **x gets close to a specific value**. It is about **getting close, not arriving**.

$$\lim_{x \rightarrow a} f(x)$$

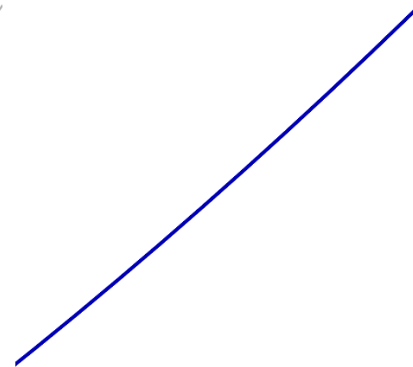
Far Away View
(Blocky/Discrete)



Zooming In
(Steps becoming a path)



The Limit
(Looks like a smooth line)



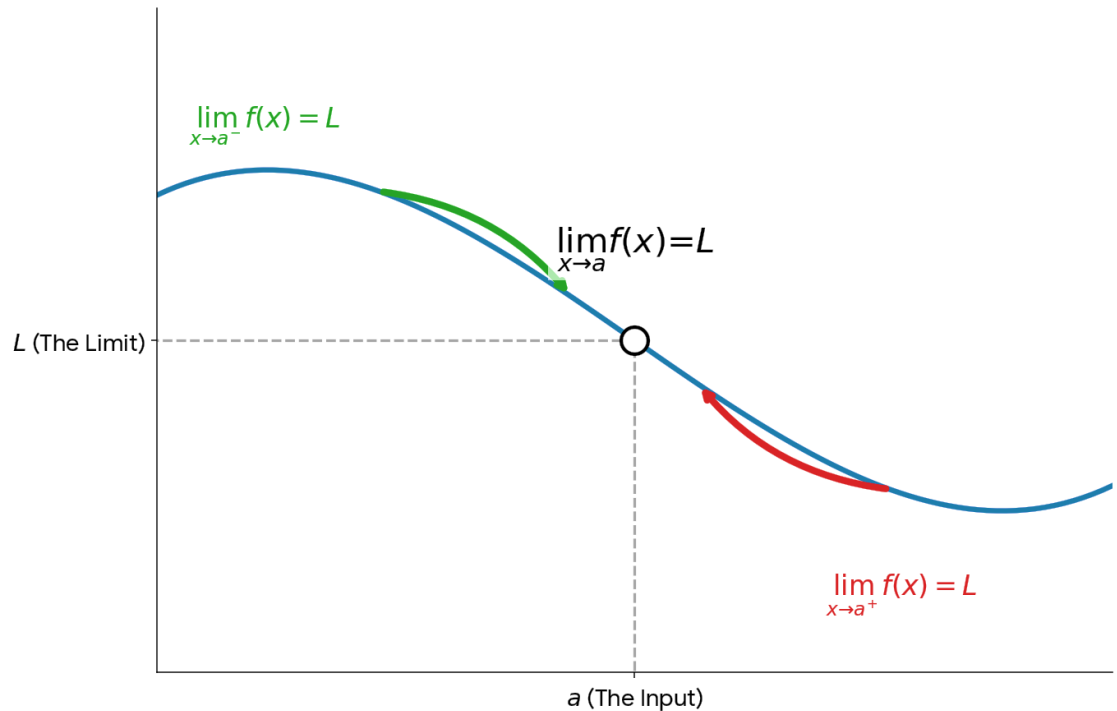
What is a Limit Graphically?

$$\lim_{x \rightarrow a} f(x) = L$$

Right and Left Hand Limits:

$$\lim_{x \rightarrow a^+} f(x) = L$$

$$\lim_{x \rightarrow a^-} f(x) = L$$

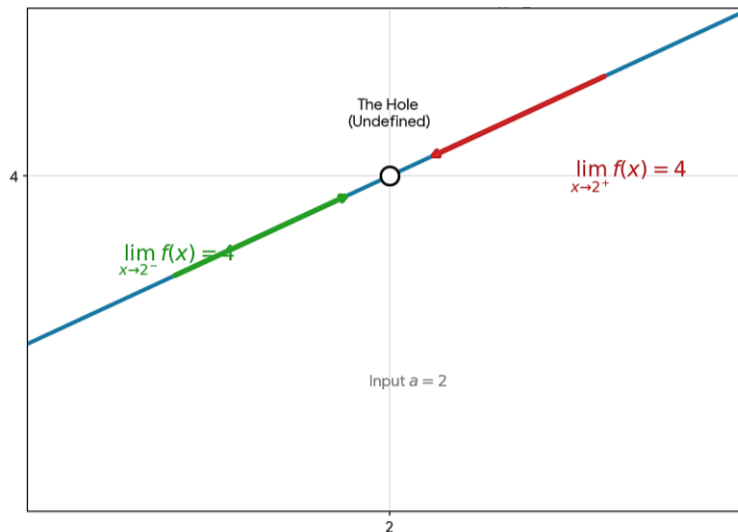


What is a Limit Numerically?

Assume:

$$f(x) = \frac{x^2 - 4}{x - 2}, x \neq 2$$

The numerical approach: $x \rightarrow 2$



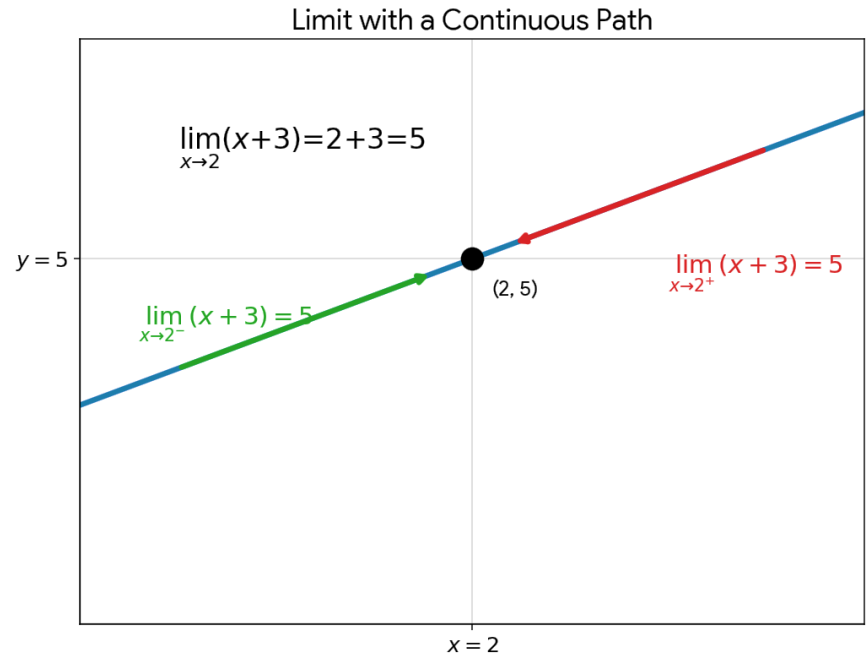
Approach	x value	$f(x)$ value
From the Left	1.9	3.9
	1.99	3.99
	1.999	3.999
The Target	2.0	Undefined (Hole)
From the Right	2.001	4.001
	2.01	4.01
	2.1	4.1



What is a Limit Algebraically?

Example:

$$\lim_{x \rightarrow 2} (x + 3) = 2 + 3 = 5$$



Question:

When can you NOT use direct substitution?

The Basic Limit Laws?

If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then:

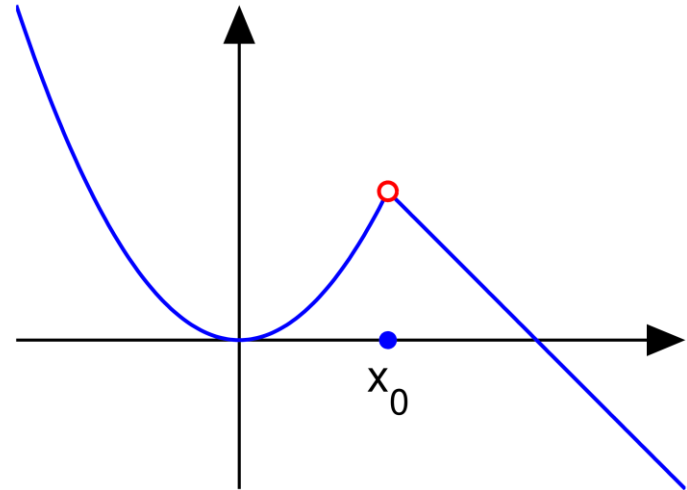
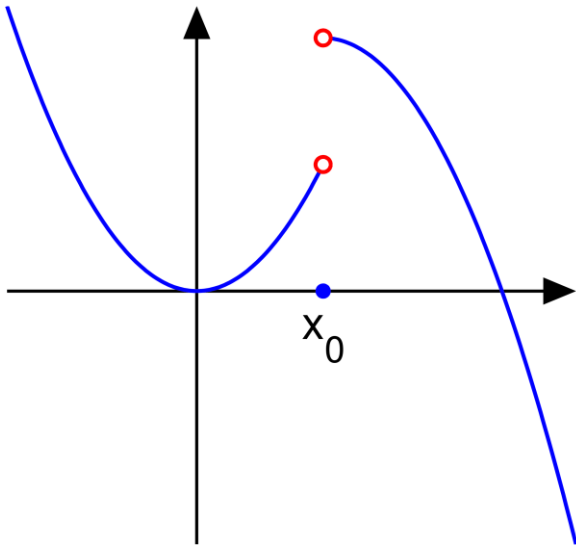
Sum Law: $\lim_{x \rightarrow a} [f(x) + g(x)] = L + M$

Product Law: $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = L \cdot M$

Quotient Law: $\lim_{x \rightarrow a} [f(x) / g(x)] = L / M$ if $M \neq 0$

Power Law: $\lim_{x \rightarrow a} [f(x)]^n = L^n$

Continuity at a Point

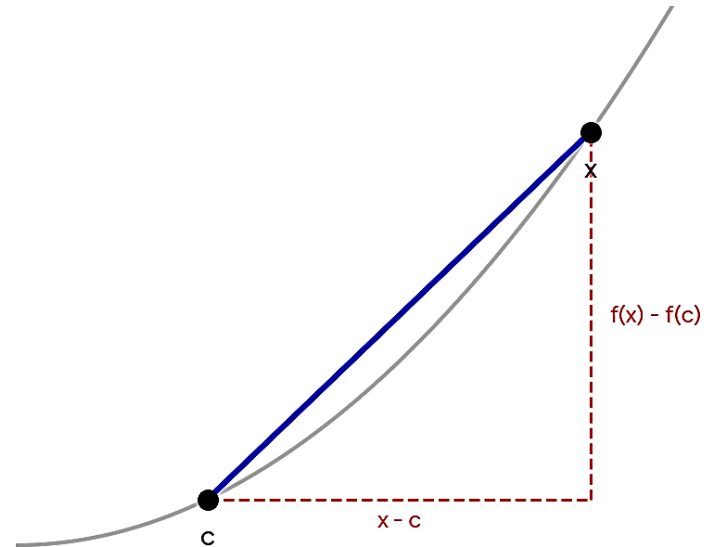


Question:
Which function is continuous?

Average Rate of Change

$$\text{Average Rate of Change} = \frac{f(x) - f(c)}{x - c}$$

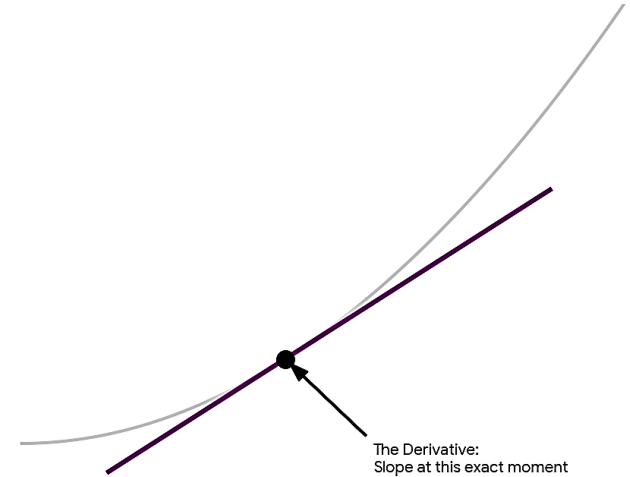
- $f(x) - f(c)$: The Change in value (the vertical "Rise")
- $x - c$: The change in time or position (the horizontal "Run")



Derivative, Instantaneous Rate of Change

The definition:

The **Derivative** is the exact rate of change at a single point.



The connection:

- **Algebraically:** it is the limit of the slope formula:

$$f'(x) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

- **Geometrically:** it is the **Slope of the Tangent Line.**

The Power Rule, Efficiency in Motion

The rule:

If $f(x) = x^n$, the **derivative** of it is:

$$f'(x) = n \cdot x^{n-1}$$

Examples: finding the slope of $f(x)$:

- $f(x) = x^3$; $f'(x) = ?$
- $f(x) = 2x^5 + 5x$; $f'(x) = ?$



Check?

What feels clear now?

What still feels uncertain?

Remember:

Calculus = Organized Common Sense
Limits → Continuity → Derivatives

Next week:

Differentiation techniques and optimization

?

Questions

- **Follow-up reading: Calculus: A Complete Course, (sections 1.2 – 1.4 & 2.2)**
- **Joining the Discussion Groups and the Lab**

